Fiscal Year:	FY 2008	**Task Last Updated:**	FY 08/07/2008
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Project Title:	The Actin Cytoskeleton: A Molecular Signature for Whole Blood Cell Cultures in the Bioreactor		
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Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Biotechnology		
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Space Biology Special Category:	None		
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Contact Monitor:		**Contact Phone:**	
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Pierson, Duane Ph.D. (NASA-Johnson Space Center) McClure, John (Baylor College of Medicine)		
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Task Description:	Studies of lymphocyte (T-cell) activation during space flight and in ground-based hypogravity model systems show a dramatic reduction in proliferative response to activating signals (Cogoli 1997; Hashemi et.al. 1999; Cooper et. al. 2001). Our Biorack flight investigations demonstrate that this inhibition of T-cell activation is accompanied by dramatic alterations in the cytoskeletal response. Specifically, activation of human peripheral T-cells by bead-immobilized anti-TCR in microgravity culture fails to induce polarization of the Microtubule Organizing Center (MTOC) towards the cell-bead contact site. These results correlate with the inability of T-cells to express surface activation markers such as the receptor for IL-2 (CD25), an event that would certainly impair the proliferative response of T-cells. More recently we have determined that exposure of the Jurkat T-cell line to hypergravity and microgravity culture can dramatically alter the state of the actin cytoskeleton and its response to stimulation. These findings have led us to the following hypothesis: The actin cytoskeleton is a gravity sensitive element of cellular response and serves as a molecular signature for gravity sensitivity of cellular function. A major goal of NASA's cellular biotechnology program is to develop new technologies for the growth of three-dimensional mammalian tissues. Tissue engineering is an emerging field with enormous potential to restore, maintain, or enhance tissues and organs (Griffith et. al., 2002). Understanding of cellular, chemical, and physical responses of cells to stimuli is critical to successfully engineering tissue. A number of studies utilize microgravity culture and Bioreactor model systems for 3-D culture of cells, cell aggregates, and tissues, such as living skin equivalents (Doolin et. al., 1999), pancreatic islets (Tobin et. al., 2001, Cameron et. al., 2001), and engineered skeletal and cardiovascular tissues (Freed et. al., 1999). While these studies are providing important data on the chemical and functional response of cells and tissues, little is known about the adaptation and response of important cellular cytoskeleton systems which play critical roles in signal transduction, functional response of cells, and tissue modeling. The actin cytoskeleton plays a crucial role in signal transduction and activation response of human cells and is involved in many aspects of cellular function including orchestration of mechanical forces inside cells, signal transduction, and functional responses of cells. The principal aim of this proposal is to employ research at the cellular level to elucidate sensitivity of the actin cytoskeleton in human peripheral blood cells (T-cells, Neutrophils, and Monocytes) in the NASA developed HARV bioreactor, and in short duration microgravity. Specific experiments will characterize the adaptation and response of the actin cytoskeleton in a number of primary human cells utilizing a novel technology we have developed to measure the polymerization state of the actin cytoskeleton in whole blood samples. The response of the actin cytoskeleton will be correlated with functional response of cellular subsets of whole blood samples (Neutrophils, T-cells, and Monocytes) cultured in (1) the HARV, (2) in short duration microgravity on KC-135 parabolic flights, and (3) in unit gravity for control. 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Scientific Significance: The interactive coupling of the cytoskeleton and signal transduction elements in regulating cellular responses is currently of extreme interest to biology and medicine. The system of microgravity-induced inhibition of cellular activation provides a unique experimental system to examine the mechanism by which the cytoskeleton and signal transduction elements regulate cellular responses. This proposal draws on results from our earlier investigations to elucidate the role of cytoskeletal systems and their interactive coupling to signal transduction elements in gravity sensitivity of cellular response. Significance to NASA: The focus of the Office of Biological and Physical Research is to foster fundamental understanding of physical, chemical, and biological processes, and building a foundation of knowledge that can be applied to Earth- and Space-based technologies. This proposal supports the goals of the Cellular Biotechnology Program by addressing key issues in cellular biology that impact cellular responses in space flight and in tissue engineering. These investigations effectively use ground-based model systems and parabolic flight microgravity experiments to enhance our understanding of fundamental biological processes to improve the understanding of environmental and gravitational influences on cells in culture. The study of whole blood cultures in bioreactors will lead to mature experiments on the space station that enable utilization of primary human cells from astronaut blood samples for studying and maintaining astronaut health. Comparison of data from parabolic flight experiments and the HARV will allow us to confirm or refute the fidelity of the ground-based bioreactor model and our hypothesis concerning the role of the actin cytoskeleton in the inhibition of cellular responses in microgravity culture. The investigations performed under this grant support the Bioastronautics Critical Path Roadmap in the Primary Risk Area of Immunology, Infection, and Hematology and they provide new insights into the suppression of cellular responses in simulated microgravity whole blood cultures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	A large number of diseases including heart disease have been associated with inflammatory responses in the past few years. As such, the interactive coupling of the cytoskeleton and signal transduction elements in regulating immune cell responses is currently of extreme interest to biology and medicine. The principal aim of the investigations performed under this grant is to understand the adaptation and response of circulating white blood cells which play a crucial role in inflamatory response. These investigations will provide new insights into the activation of inflamatory cells in whole blood cultures.

Task Progress:	In year one of this grant we developed a novel method for establishing whole blood cell cultures in the rotating HARV. A large number of experiments were devoted to evaluating various culture conditions and developing new methods for Autologous Plasma Supplementation of whole blood cultures. These investigations enabled us to develop superior culture environments for whole blood by increasing the buffering capacity of blood cultures in a simulated microgravity environment. In year two and three of this grant we utilized the methods developed in year one to characterize the adaptation and response of circulating white blood cells to modeled microgravity, and their re-adaptation to 1-g upon return to static culture. In addition, we developed new methods for characterizing infection of human blood cells upon bacterial infection. These later investigations demonstrate the utility of this new method for identification of human blood infections and may be developed for advanced monitoring tool during long duration spaceflight. In summary, the investigations performed under this grant have provided deep insight into the adaptation and response of human circulating blood cells using the HARV model system. The technologies developed during this grant are providing new tools for monitoring immune cells and their functional responses in whole blood samples. The use of Actin signatures on long-duration space flight missions can provide early detection of cellular damage in flight crew prior to clinical manifestations. Earth-based applications of this technology include new diagnostic tools for the clinical assessment of human immune cells and monitoring of disease evolution in patients.
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